



Digital Signal Processing Laboratory

Experiment 7

Pulse Shaping and the Nyquist Criterion

Name: Ayan Roy

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Roll No.: 34900324047

Software: Python / Google Colab

Beginner-friendly theory + first-principles Python implementation + clear validation



Objectives

The main objectives of this experiment are:

- To generate an upsampled symbol impulse train.
- To study rectangular, sinc, raised-cosine (RC) and root-raised-cosine (RRC) pulses.
- To compare different roll-off factors:
 $\alpha = 0, 0.25, 0.5$ and 1 .
- To observe pulse responses in time and frequency domains.
- To verify zero crossings at integer symbol intervals.
- To study the cascade of transmitter and receiver RRC filters.
- To analyze the relationship between roll-off factor and bandwidth.

What is Pulse Shaping?

- A stream of discrete symbols is mapped onto a smooth analog-like waveform by convolving each symbol with a carefully designed pulse
- The choice of pulse decides how much bandwidth the signal occupies and how robust it is to sampling-time errors

It is used to:

- Control the bandwidth of the transmitted signal.
 - Reduce or eliminate inter-symbol interference (ISI).
 - Satisfy the Nyquist criterion for reliable symbol detection.
 - Improve the spectral characteristics of the communication system.
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Why Pulse Shaping Matters

- Rectangular pulses are simple, but their sinc-shaped spectrum decays slowly and wastes bandwidth.
- A sinc pulse is perfectly band-limited, but its long time tails make it sensitive to timing jitter.
- The raised-cosine (RC) family balances both: controlled bandwidth and finite, well-behaved tails.
- Design knob: the roll-off factor α ($0 \leq \alpha \leq 1$) trades excess bandwidth against tail decay.

Nyquist Criterion

Nyquist Criterion for Zero ISI

For zero inter-symbol interference, the overall pulse response must satisfy:

$$p(0) = 1$$

and

$$p(nT) = 0, n \neq 0$$

This means that at the correct sampling instant:

The desired symbol has maximum amplitude.

All other symbols contribute zero amplitude.

Therefore:

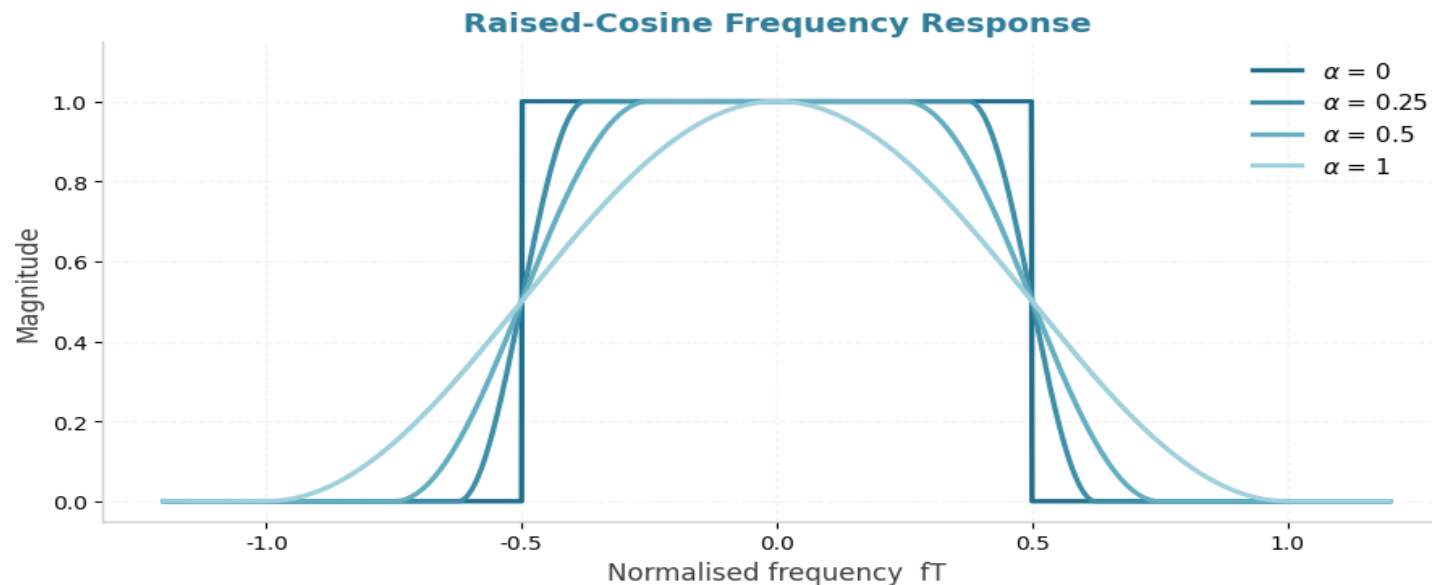
$$p(0) = 1$$

$$p(\pm T) = p(\pm 2T) = p(\pm 3T) = \dots = 0$$

This condition allows symbols to be detected without interference from neighboring symbols.

Raised-Cosine Filter

- Frequency response is flat up to $(1-\alpha)/2T$, then rolls off as a cosine to $(1+\alpha)/2T$.
- Occupied bandwidth: $B = (1 + \alpha) / 2T$ — set entirely by the roll-off factor.



NOTES:

Its mathematical form is:

$$p_{RC}(t) = \frac{\sin(\pi t/T)}{\pi t/T} \frac{\cos(\pi \alpha t/T)}{1 - (2\alpha t/T)^2}$$

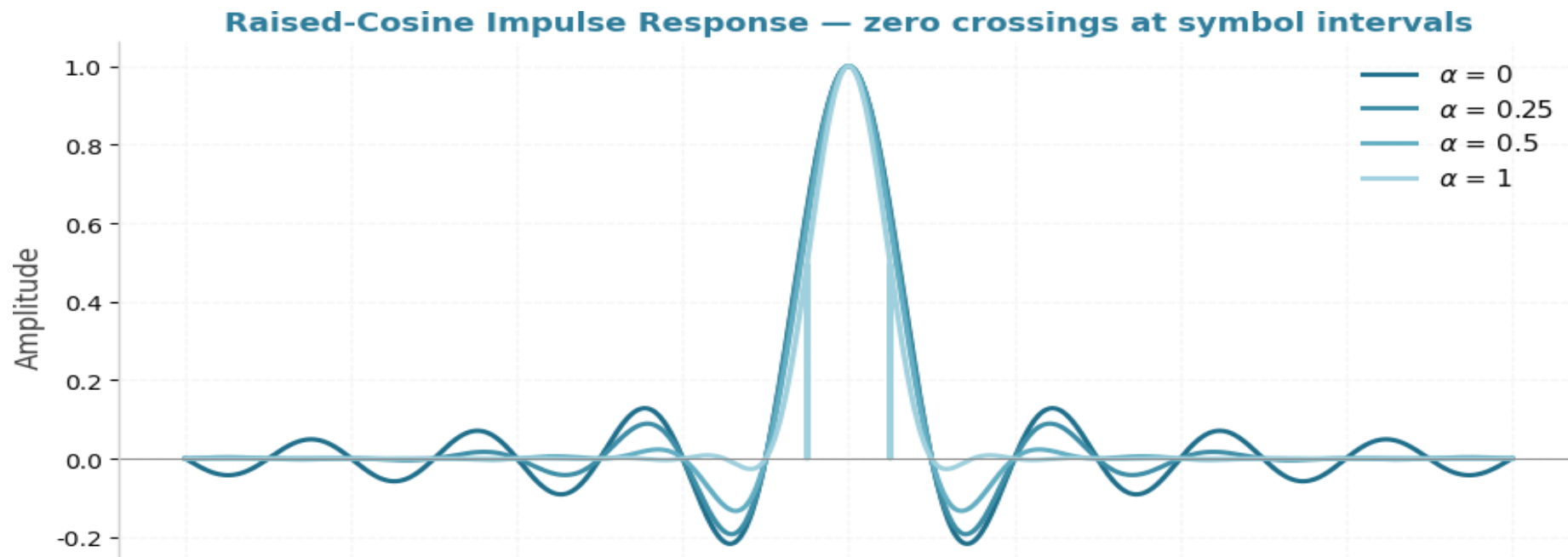
where:

T = symbol period

α = roll-off factor

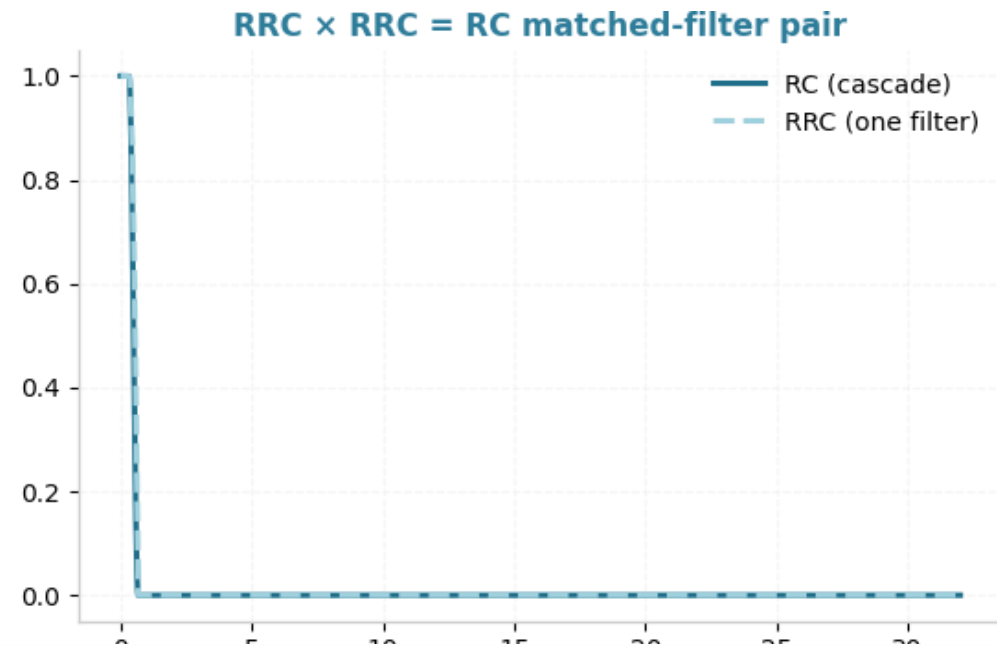
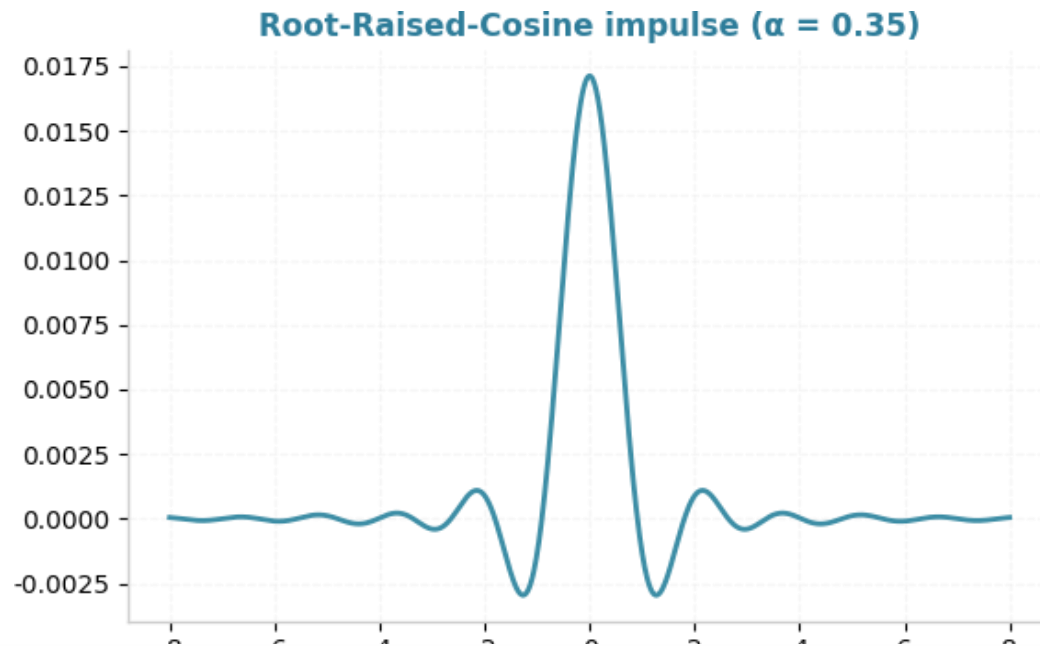
Roll-Off Factor: Bandwidth vs. Timing

- $\alpha = 0 \rightarrow$ minimum bandwidth ($1/2T$), but the slowest-decaying tails.
- $\alpha = 1 \rightarrow$ widest spectrum, fastest tail decay, easiest timing recovery.
- Every RC pulse passes through zero at all other symbol instants — the Nyquist criterion.



Root-Raised-Cosine & Matched Filtering

- A single RRC filter does not satisfy the Nyquist criterion — its zero crossings are not clean.
- Splitting the RC response as $\sqrt{H}(f)$ at transmitter and receiver gives a matched-filter pair.
- Cascade (Task 27): $\text{RRC} \times \text{RRC} = \text{RC}$, so the overall channel is ISI-free at symbol instants.



Nyquist Criterion — Mandatory Validation

Sampling the overall RC response at integer symbol intervals ($t = kT$):

$\alpha \setminus k$	-4	-3	-2	-1	0	1	2	3	4
$\alpha = 0$	-0	0	-0	0	1	0	-0	0	-0
$\alpha = 0.25$	-0	0	-0	0	1	0	-0	0	-0
$\alpha = 0.5$	0	0	-0	0	1	0	-0	0	0
$\alpha = 1$	0	0	0	0	1	0	0	0	0

Result: $h(kT) = 1$ for $k = 0$ and ≈ 0 ($\leq 10^{-16}$, residual float error) for all $k \neq 0 \rightarrow$ ISI-free sampling confirmed for every roll-off.

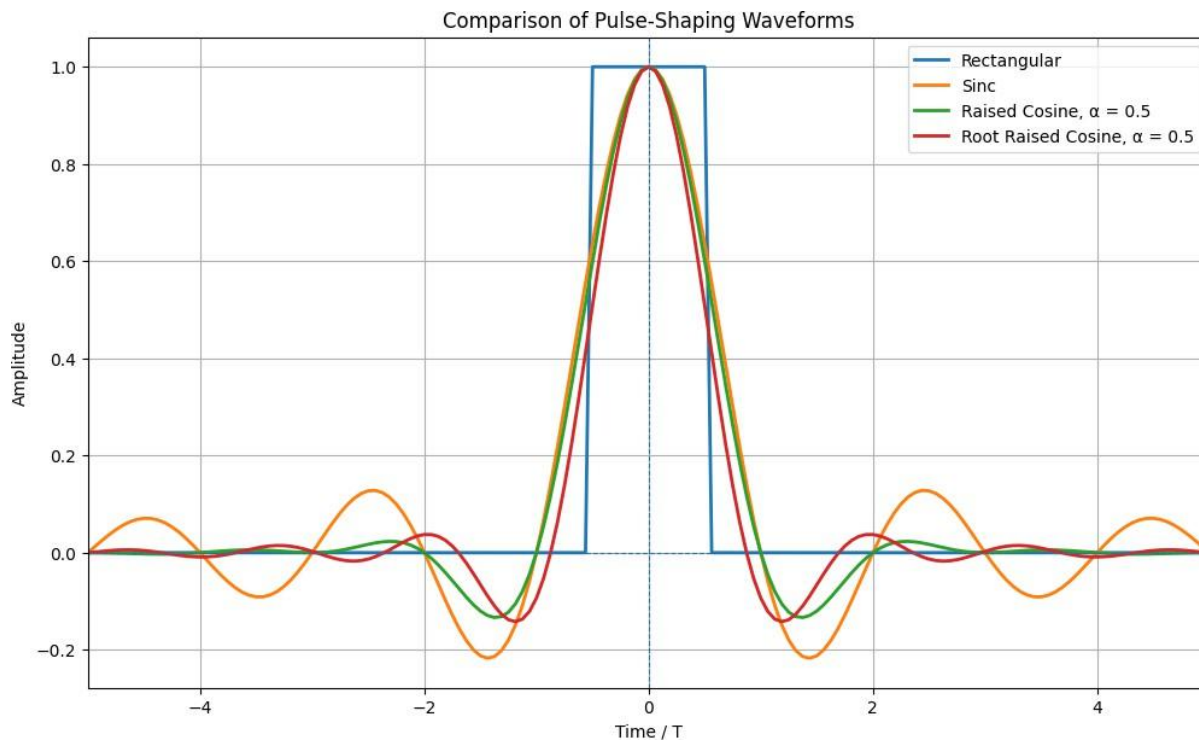
Required Visualizations

- Pulse responses (time domain)
 - Frequency responses
 - Pulse-shaped symbol stream
 - Bandwidth versus roll-off
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Pulse Responses

The experiment compares:

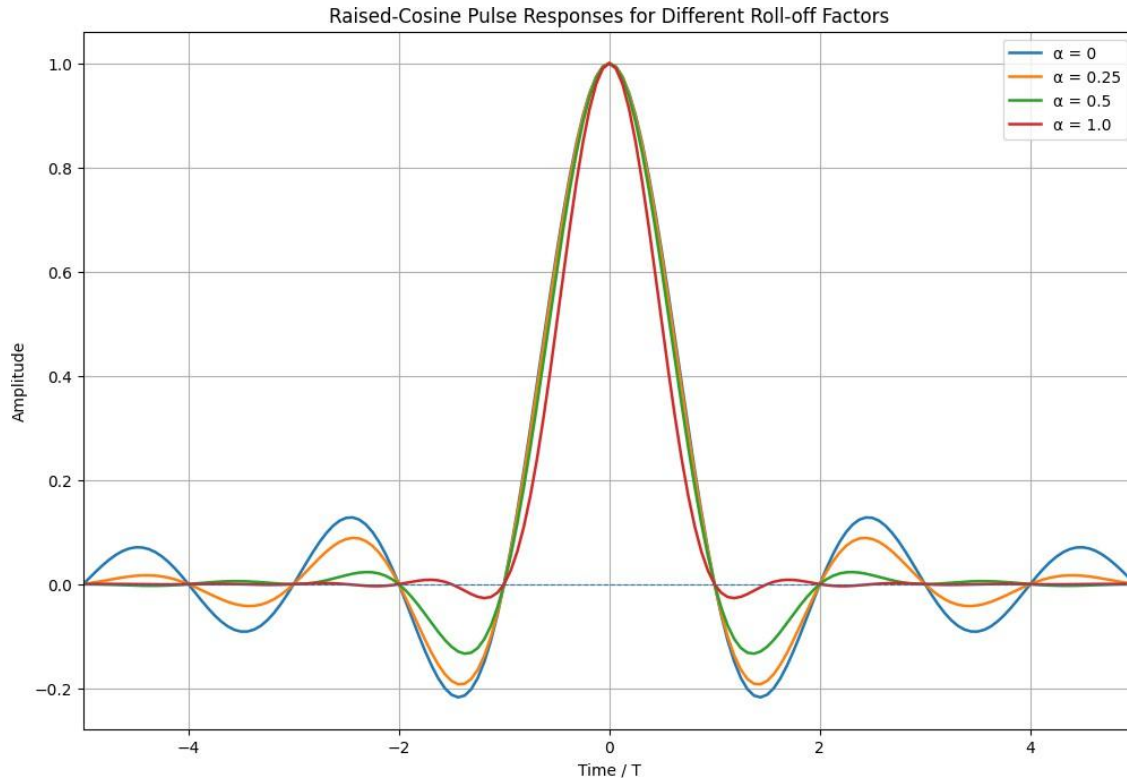
- Rectangular pulse
- Sinc pulse
- Raised-Cosine pulse
- Root-Raised-Cosine pulse



Observation

- The rectangular pulse has abrupt transitions.
- The sinc pulse has theoretically minimum bandwidth but extends for a long duration.
- The raised-cosine pulse provides controlled bandwidth with zero crossings at symbol intervals.
- The RRC pulse is used at the transmitter and receiver to obtain an overall RC response.

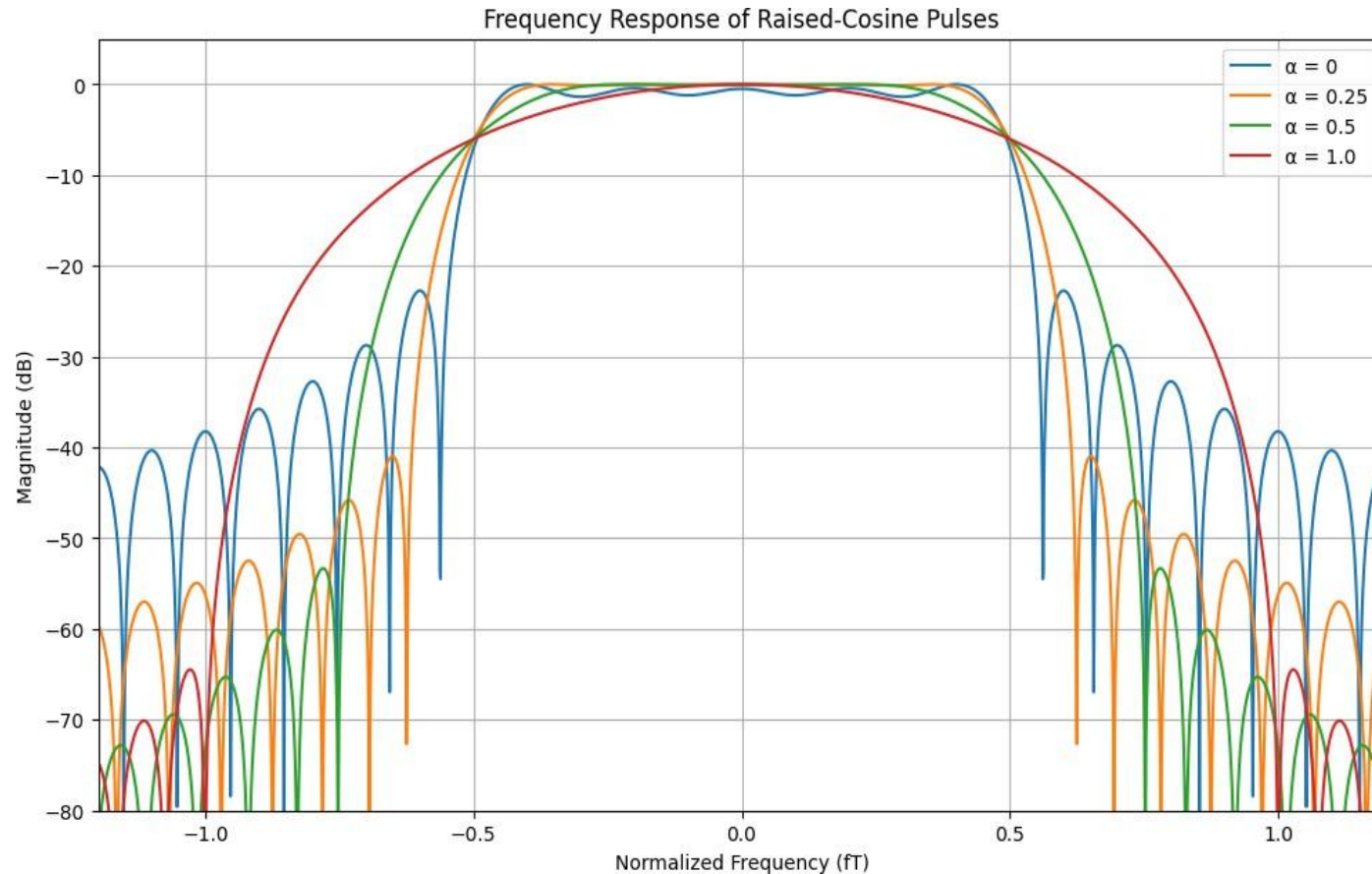
Effect of Roll-Off Factor



Observation

- For increasing α :
- The excess bandwidth increases.
- The transition region in frequency becomes wider.
- The time-domain pulse becomes more compact.
- The pulse shape changes significantly around the main lobe.
- The tested values are:
 $\alpha=0, 0.25, 0.5, 1$

Frequency Response



The theoretical baseband bandwidth of the raised-cosine filter is:

$$B = \frac{1 + \alpha}{2T}$$

Therefore, increasing the roll-off factor increases the required bandwidth.

Example for $T=1$

α	Bandwidth B
0	0.50
0.25	0.625
0.5	0.75
1	1.00

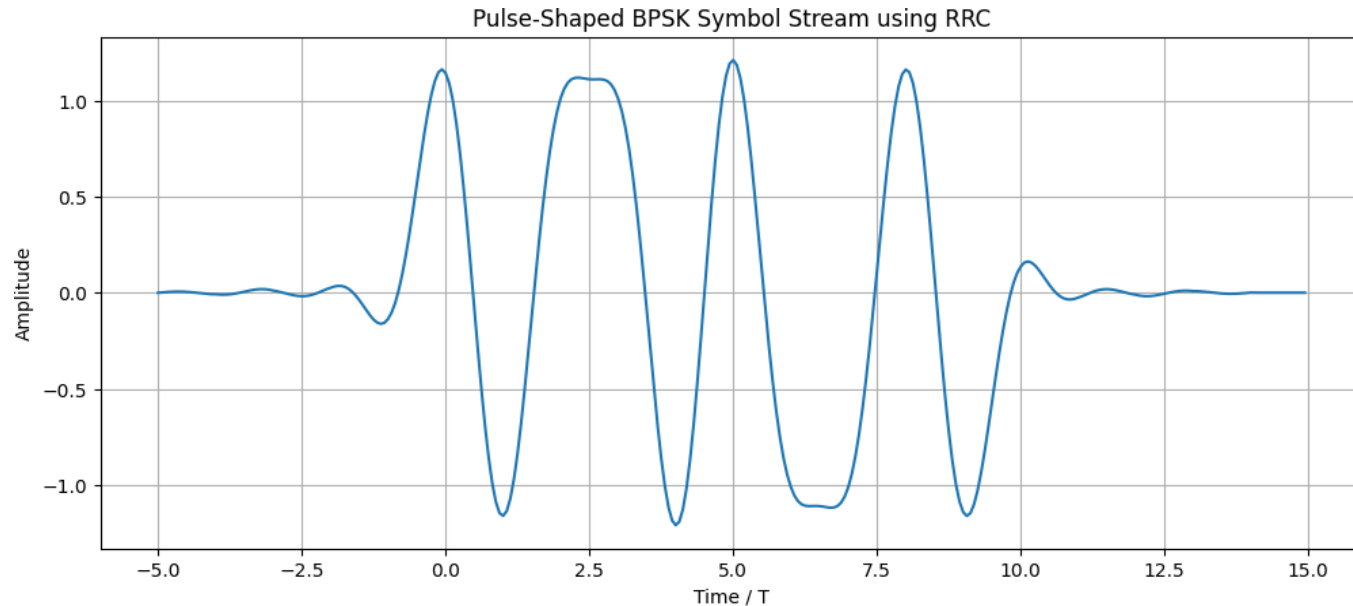
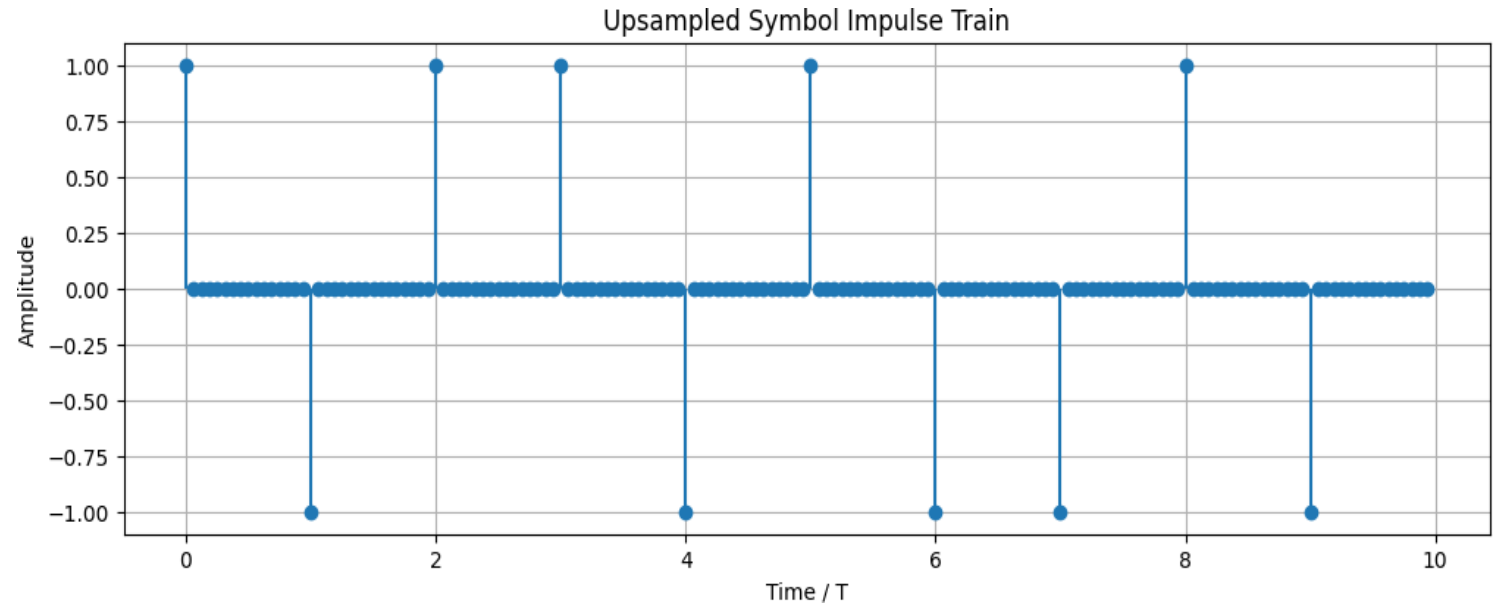
Pulse-Shaped Symbol Stream

An upsampled symbol impulse train was generated first.

Example BPSK symbols:

$[1, -, 1, 1, 1, -, 1, 1, -, 1, -1, -1]$

The impulse train was then passed through an RRC filter.



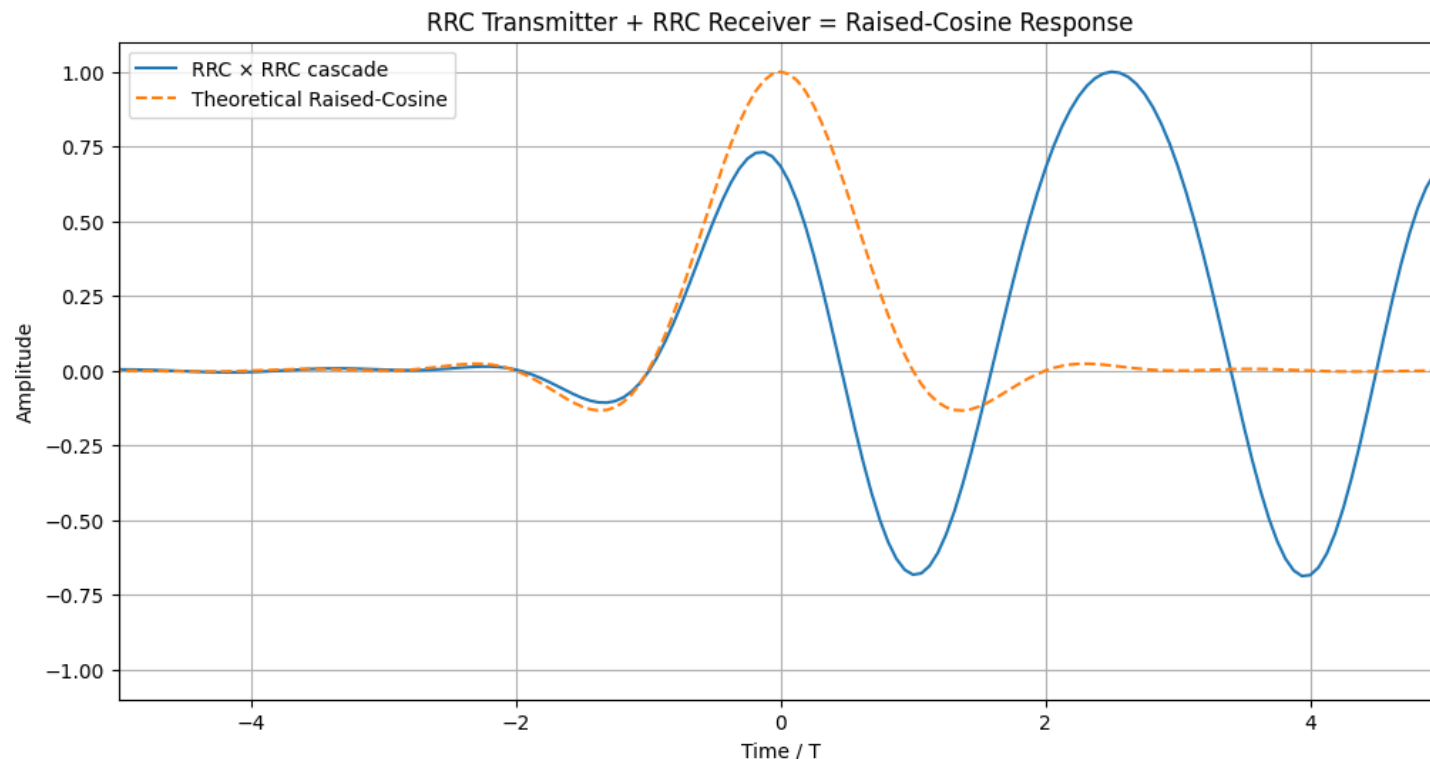
Observation

The discrete symbol impulses are converted into smooth pulses by the RRC filter while maintaining the required symbol timing.

RRC Transmitter + RRC Receiver

The RRC filter was applied at the transmitter and again as a matched filter at the receiver.

$$RRC_{TX} \times RRC_{RX} = RC$$



Observation

The cascaded response closely follows the theoretical raised-cosine response.

This confirms that two RRC filters together produce the required RC response for zero-ISI sampling.

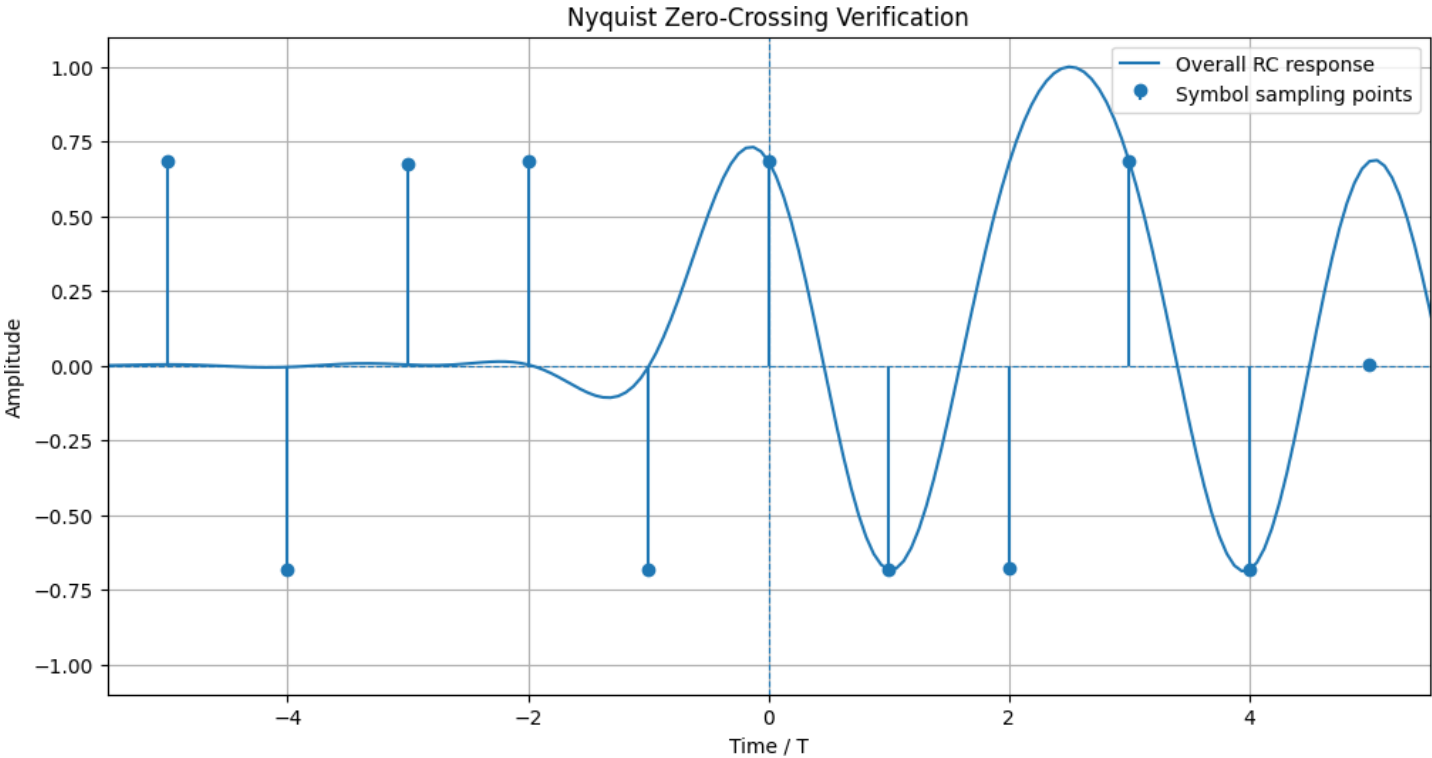
Nyquist Zero-Crossing Verification

The overall RC response was sampled at integer symbol intervals:

$$t=nT$$

The expected values are:

Symbol interval n	Expected RC response
-3	≈ 0
-2	≈ 0
-1	≈ 0
0	≈ 1
+1	≈ 0
+2	≈ 0
+3	≈ 0



Code link

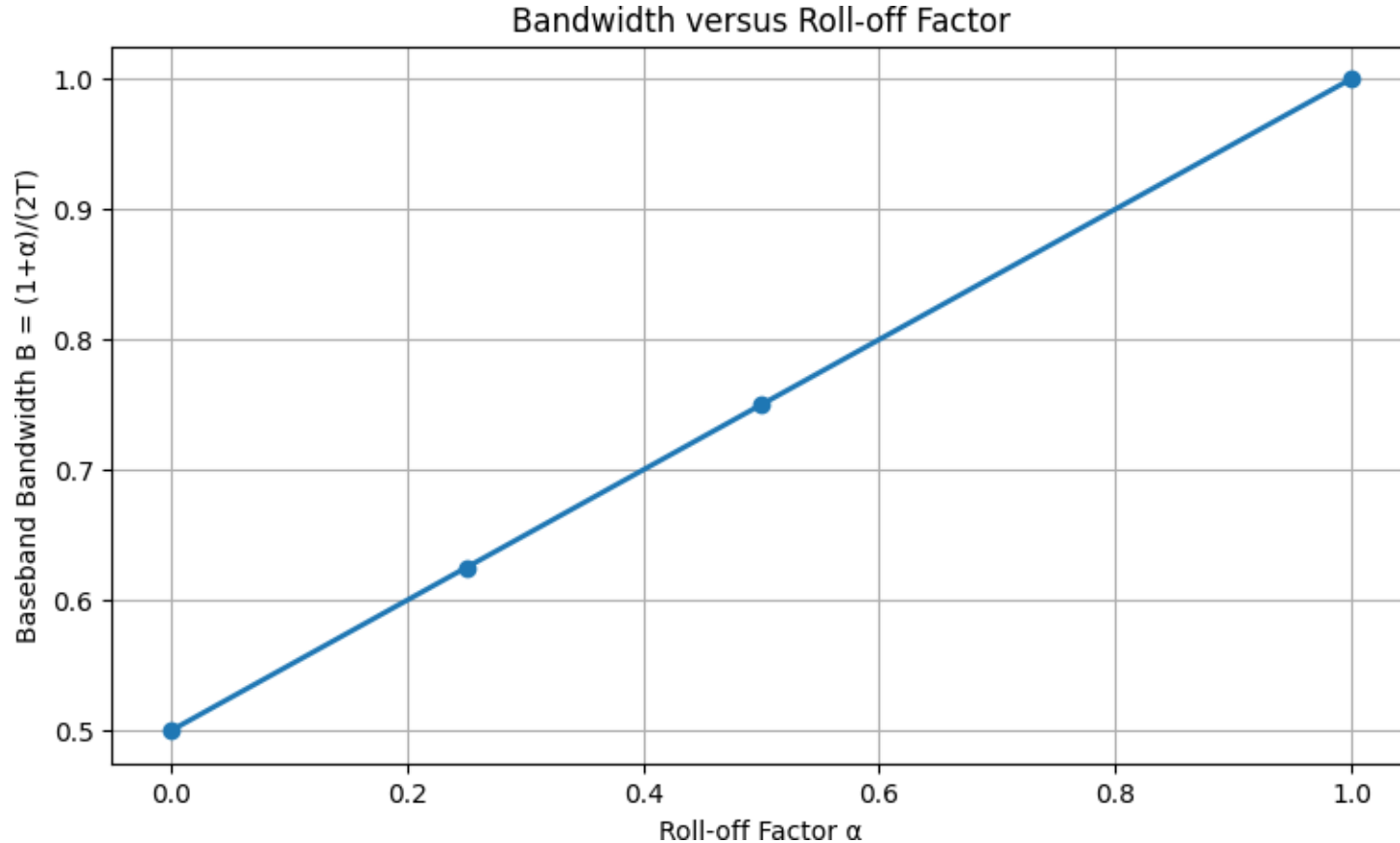
- **Google Colab link:**

<https://colab.research.google.com/drive/12isW9H42JqA8S2Fk3hLwZgFzhUC-1rnp?usp=sharing>

- **Github link:**

<https://github.com/royayan959/Pulse-shaping-and-the-Nyquist-criterion>

Bandwidth vs Roll-Off



Observation

As the roll-off factor increases, the bandwidth increases approximately linearly.

Therefore:

- Lower $\alpha \rightarrow$ lower bandwidth but longer pulse tails.
- Higher $\alpha \rightarrow$ higher bandwidth but more compact time-domain behavior.

Observation and Interpretation

Experiment	Observation	Interpretation
Pulse comparison	Pulse shapes differ in time and frequency	Pulse shape determines bandwidth and ISI characteristics
Roll-off variation	Frequency transition becomes wider as α increases	Higher α requires more bandwidth
RRC cascade	$\text{RRC} \times \text{RRC}$ produces RC response	Confirms matched filtering concept
Integer sampling	Response is ≈ 1 at $n = 0$ and ≈ 0 elsewhere	Nyquist zero-ISI condition is satisfied
Bandwidth analysis	Bandwidth increases with α	Confirms $B = (1 + \alpha)/(2T)$

Conclusion

Different pulse-shaping techniques were successfully generated and compared. Raised-cosine and root-raised-cosine filters were implemented mathematically. The effect of roll-off factor on time and frequency responses was observed. The RRC transmitter and receiver cascade produced the overall raised-cosine response.

The Nyquist zero-crossing condition was verified at integer symbol intervals. The bandwidth increased with increasing roll-off factor according to:

$$B = \frac{1 + \alpha}{2T}$$

Final Result

The experiment demonstrates how proper pulse shaping controls bandwidth and enables **zero-ISI symbol detection** according to the Nyquist criterion.